Adam Sirota, Alex Zheng - Assignment 1 Writeup

Dataset and goal

This visualization explores daily weather measurements from U.S. weather stations during 2017. The data comes from the NOAA Daily Global Historical Climatology Network dataset provided for the assignment. We focus on two variables: average daily maximum temperature and average daily precipitation, as temperature and precipitation are important indicators of a location's climate and help characterize differences in weather patterns across regions.

Our goal was to help users explore how weather varies geographically and discover how temperature and precipitation correlate. For example, does it rain more in colder or warmer locations, and where are these locations on the map? After cleaning null temperature values and combining daily records grouped by station, the visualization contains 1,559 stations across the US. Average daily high temperatures across 2017 range from approximately 24.7°F to 93.7°F, while average daily precipitation ranges from 0 to 0.539 inches.

Design rationale

We used two coordinated views: a geographic map of the US and a scatterplot. The map places each station in its geographic location, making regional patterns easy to identify and providing a familiar interface to users. We included a basemap to provide geographic context and help users identify the state boundaries within which stations are located. Meanwhile, the scatterplot reveals relationships and trends between temperature and precipitation.

Stations are colored using a sequential yellow/red scale for average daily maximum temperature, with darker red indicating hotter locations. We chose this as the light-to-dark progression conveys increasing temperature and red is commonly associated with heat. We also considered using alternative color scales like rainbow or diverging, but rejected rainbow because hues have no natural order and it creates false bands and the diverging scale because a data-derived midpoint isn't a meaningful threshold. A legend was also added to both the map and scatterplot so users can interpret the colors and estimate a station's temperature. Using the same color encoding in both views helps users connect a station's temperature to its location and position in the scatterplot.

By hovering a station on either visual, we can see the corresponding station in the other visual. We also have added presets to capture the top 10% hottest, coldest, and wettest environments, which allows for easy exploration of extreme values to gain more insight on how extreme temperature and precipitation clusters in certain regions. In combination with the linking, we can identify these clusters in both the scatterplot and the map to visualize the trends and geographic patterns. For example, the hottest stations cluster in the South and Hawaii, the coldest follow the Rocky Mountains and Alaska, and the wettest stations appear in two distinct regions: the Southeast and the Northwest. We also annotated the hottest, coldest, and wettest stations on the scatterplot so the extremes are easy to find, and hovering any of them highlights its location on the map through the hover link.

We added both pan and zoom as interactive features with a button toggle instead of an alternative like a toggle key on the keyboard which could be difficult to find. As both the brush and pan need a dragging motion, this was the easiest way to allow for both features without issues with

drag interactions. The toggle also makes it clearer when users are navigating the map rather than interacting with individual stations.

Interactions

The visualization includes the following interactions:

Brushing and linking (Category I): Users can drag a rectangular selection area on either the map or scatterplot, which can be resized and moved. The selected stations remain visible while non selected stations are muted in both views. Also, the selected stations in one plot are highlighted in the other one. A label above the button menu updates after brushing selections to reflect the number of selected stations and their average temperature/precipitation numbers.

Preset selections (Category I): There are buttons on a menu above the graphs in which the users can select and highlight the top 10% hottest, coldest, or wettest stations on the graphs. You can click in an empty space or use the clear selection button in order to reset the selection to an empty state.

Details on demand (Category II): Hovering over a station on either graph displays its name, state, average high temperature, and average daily precipitation.

Hover linking (Category II): Hovering a station on either chart outlines the same station in the other chart, so users can find a single station in both views at once.

Pan and zoom (Category II): Users can select the pan and zoom mode on the button menu to enable these actions for the map, and then click the reset view button when they are finished. Use

the scroll wheel while in this mode to zoom into the map and drag to pan around the zoomed map. Switching back to the selection mode will let users brush the visible stations. While panning, brushing is unavailable, and you can see by the cursor which mode you are in: the crosshair cursor is selection mode, and the hand cursor is pan mode.

Annotation (Category II): The scatterplot points out the hottest, coldest, and wettest stations. By hovering any of these stations in the scatterplot, the corresponding station in the map is also highlighted.

Development process

Adam worked on the baseline page, cleaning the data, adding the initial map and scatterplot, added basic styling, and also added the brushing and linking features. Alex added the basemap, the temperature color scale and legends, linking upon hovering over a station, the button menu and respective functions, including the preset selections and pan&zoom, the annotations, and other small touchups like filtering for just US states.

In total, developing this project took about a combined 12 hours across the two of us, with an additional 2 to 3 hours spent on the writeup, demo video, and publishing the site. The aspects that took the most time were figuring out how to best implement features that were not yet covered in class, which includes some of the features listed in categories 1 and 2. Integrating these features while ensuring that they worked together without conflicting required experimentation and debugging.

AI usage

AI tools were used to find guidance on implementing visualization features, suggest implementation approaches, assist with debugging, and review and refine the project write-up.

References

NOAA Daily Global Historical Climatology Network weather data, provided as weather.csv for CMSC 471 Assignment 1.

D3.js for data loading, scales, SVG rendering, brushing, zooming, and tooltips.

TopoJSON Client for rendering the provided U.S. basemap.